# HARISHALINEE ELANGOVAN

**Software, Data & AI Professional | .NET | Python | SQL | Automation Testing**
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## PROFESSIONAL SUMMARY

Associate Data Analyst & AI Engineer and BSc (Hons) Information Technology (Data Science) undergraduate at SLIIT, with hands-on experience in full-stack .NET Framework development, REST APIs, IIS deployment, Docker, OCR and document processing, automation testing, data pipelines, interactive dashboards and machine learning. Built an edge-based fall-risk detection model with 97.12% accuracy on unseen subjects and co-authored a paper accepted for ICSCDS-2026 (IEEE proceedings). Proficient in Python, SQL, Power BI, TensorFlow, PyTorch, AWS and Azure. Open to Software Engineer, Data Analyst, AI/ML Engineer and QA Automation roles.

## TECHNICAL SKILLS

- **Core Competencies:** Software development, data analytics, business intelligence, data visualization, data pipelines, ETL, machine learning, deep learning, NLP, computer vision, test automation, application deployment, containerization, OCR, document processing, cloud computing
- **Programming & Frameworks:** Python, SQL, .NET Framework (back-end and front-end), REST APIs, Java, C/C++, R, FastAPI, Flask, React
- **Data Analysis & BI:** Power BI, SSRS, Advanced Excel, Matplotlib, Seaborn, Streamlit, Monday.com, ETL, SSIS, data warehousing, OLAP (SSAS), data modeling, Oracle, SQL Server
- **AI & Machine Learning:** Machine learning, deep learning, Scikit-Learn, XGBoost, TensorFlow, PyTorch, NLP, computer vision (MediaPipe), LangChain, LangGraph, Gemini API, generative AI (StyleGAN)
- **Cloud, Testing & Deployment:** AWS, Microsoft Azure, IIS, Docker, application deployment, automation testing, Git, GitHub
- **Professional:** Problem-solving, stakeholder management, information systems auditing, controls and risk assessment

## WORK EXPERIENCE

**Associate Data Analyst & AI Engineer | Agrithmics | Feb 2026 – Present**
- Develop back-end and front-end features in .NET Framework and build REST APIs
- Deploy applications on IIS, work with Docker containers, and release data/AI solutions to live environments
- Build OCR-based text extraction and document-handling features
- Carry out automation testing to verify that dashboards, data pipelines and reporting workflows work correctly
- Led development of multiple interactive dashboards for plantation performance tracking across 6–7 projects
- Optimized data pipelines that feed AI model training, improving processing efficiency
- Built automated reporting workflows that speed up decision-making

**Data Analytics & AI Development Intern | Agrithmics | Nov 2025 – Jan 2026**
- Designed interactive dashboards to visualize plantation performance
- Preprocessed and structured large datasets for AI model training and optimization

**Data Analyst Intern | International Youth Alliance for Peace (IYAP) | Mar 2025 – Oct 2025**
- Used Monday.com to manage data, build dashboards and track project performance
- Analysed data and produced reports to support strategic planning

**IT Operator | Crown Global International Pvt Ltd | Jul 2024 – Feb 2025**
- Managed digital record-keeping systems and ensured accuracy of computerized accounts
- Performed system maintenance and supported branding operations through media coordination

**Freelance Tutor | Self-employed | Jan 2023 – Present**
- Provide personalized tutoring sessions that improve students' understanding and academic performance

## PROJECTS

**Edge-Enabled Skeletal Movement Analysis System for Real-Time Fall Risk Detection | Final-year research (CareSense group project R26-DS-010) | 2026**
- Built a privacy-preserving fall-risk detector that runs on an ordinary RGB webcam and a CPU, analysing skeleton data only
- Fused an ST-GCN (PyTorch, MediaPipe) with 18 biomechanical features in a late-fusion model, trained on NTU RGB+D and UR Fall Detection (578 labelled clips)
- Achieved **97.12% accuracy, 92.31% sensitivity, 1.28% false-alarm rate and ROC-AUC 0.9836** on unseen subjects, with 4.5 ms inference per window on a CPU
- Measured domain shift (cross-dataset ROC-AUC 0.410) and recovered to macro-F1 0.8851 on webcam data using 35 fine-tuning clips

**Tourism AI – Multi-Agent Travel Optimization System | Python, Gemini API, LangGraph, LangChain, FastAPI, React**
- Engineered a multi-agent system that builds personalized, Sri Lanka-aware travel itineraries
- Designed specialized agents for interest understanding, day planning, weather-aware re-planning, transport and food tips
- Delivered as a scalable FastAPI and React full-stack application

**Caregiver Stress Prediction Dashboard – Wearable Sensor Analytics (Group Project) | Python, FastAPI, Streamlit, SQLite, Plotly, Matplotlib, Seaborn, LLM chatbot**

- Built an interactive analytics dashboard with a FastAPI back end and SQLite database that predicts and visualizes caregiver stress levels (Low / Medium / High) from wearable sensor data
- Responsible for the movement-magnitude analysis in a 4-member team, using coordinated Plotly, Matplotlib and Seaborn charts to show stress patterns
- Integrated an LLM-powered chatbot that explains trends and highlights unusual stress patterns for healthcare managers

**Ride-Hailing AI – Data Mining and Prediction System | Python, Scikit-Learn, Streamlit**

- Mined about 148,000 ride bookings using Apriori / FP-Growth route mining and K-Means / DBSCAN rider segmentation
- Built models to predict high-demand windows and forecast monthly spend, and recommended money-saving subscription plans in an interactive Streamlit dashboard

**Customer Churn Prediction – Telco Churn Lab | Python, Scikit-Learn, XGBoost, Flask, React, Vite**

- Trained and compared Random Forest and XGBoost classifiers on 7,043 customers using accuracy, precision, recall and F1; Random Forest performed best
- Shipped a full-stack app with a Flask prediction API and a React + Vite front end

**Hotel Booking Data Warehouse and BI | SQL Server, SSIS, SSAS, Power BI, SSRS**

- Designed a star schema with a booking fact table and six dimensions, built SSIS ETL packages and an SSAS OLAP cube
- Reported booking trends, revenue and cancellation rates in Power BI and SSRS dashboards

**StyleGAN and StyleGAN2 – Bias and Fairness Analysis | PyTorch, StyleGAN2-ADA, OpenCV**

- Analysed how StyleGAN2 models trained on FFHQ favour lighter skin tones and Western faces, and proposed a South Asia-aware fairness framework using ITA skin-tone metrics

## RESEARCH PUBLICATION

**Multimodal AI Framework for Intelligent Caregiver Support and Patient Safety in Care Facilities**
Dayana Priyadharshani Kumar, Harishalinee Elangovan, Vidursha Prabagaran, Kaushalya Nagenthiraraja, Samadhi Rathnayake, Samantha Thelijjagoda. 4th International Conference on Sustainable Computing and Data Communication Systems (ICSCDS-2026), IEEE conference proceedings, 15–17 December 2026, India. Status: accepted for presentation and publication (acceptance with major revision), Paper ID ICSCDS-0291.
CareSense research portfolio (project documents, results and presentations): caresense-portfolio.vercel.app

## EDUCATION

**BSc (Hons) in Information Technology, specializing in Data Science | SLIIT | Jan 2025 – Mar 2027 (expected) | CGPA 3.25 / 4.00**

**Higher Diploma in Information Technology | SLIIT | Oct 2022 – Nov 2024**

**Cambridge Advanced Level | Lyceum International School, Sri Lanka | 2020 – 2022 | 1 A*, 1 A, 1 C**

**Cambridge Ordinary Level | Lyceum International School, Sri Lanka | 2010 – 2020 | 2 A*, 2 A, 2 B, 1 C**

## CERTIFICATIONS AND COURSES

- AI/ML Engineer (Stage 1 and 2) – The Centre for Open and Distance Education, Faculty of Computing, SLIIT
- AWS Cloud Computing Certifications – AWS
- Microsoft Azure Cloud Certifications – Microsoft
- Information Systems Auditing, Controls and Assurance – Coursera
- Oracle Database 19c; Introduction to Artificial Intelligence; R for Data Science: Analysis and Visualization; Python Data Analysis; Getting Started with Power BI – LinkedIn Learning

## PERSONAL DETAILS AND LANGUAGES

Date of birth: 6 October 2004 | Nationality: Sri Lankan | Languages: English, Tamil, Sinhala

## REFERENCES

**Mrs Laurel Mathew**, Director, Crown Global International (Pvt) Ltd | 0777788969 | mathewlaurel@gmail.com
**Dr. Samantha Rajapaksha**, Head of IT Department and Senior Lecturer (HG), SLIIT | 0117544802 | samantha.r@sliit.lk